

National Neighborhood Data Archive (NaNDA): Retail Establishments by Census Tract, United States, 2003-2017



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nanda_retail_tract_2003-2017_01P.dta

nanda_retail_tract_2003-2017_01P.csv

nanda_retail0317T_01P.sas7bdat

Overview and Data Dictionary

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Dataset Overview

Description

This dataset contains measures of the number and per capita density of select types of retail establishments—such as clothing, department, building and garden, furniture, and thrift stores—per United States census tract from 2003 through 2017.

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Data Sources

The North American Industry Classification System (NAICS) was used to identify organizations of interest. NAICS was developed by the United States Office of Management and Budget and is “the standard used by Federal statistical agencies in classifying business establishments for the purpose of collecting, analyzing, and publishing statistical data related to the U.S. business economy” (United States Census Bureau, 2017).

Establishment data was taken from the National Establishment Time Series (NETS) database, which provides annual records of the US economy for more than 60 million private for-profit and nonprofit establishments and government agencies (Walls & Associates, 2017). This longitudinal data source classifies establishments by NAICS code and provides detailed business microdata and address history. NETS is produced in most years by Walls and Associates using archival establishment data from Dun & Bradstreet. While it currently includes data from 1990 to 2015 and 2017, only data from 2003-2015 and 2017 were used to create this dataset. (NETS data was not produced for 2016.)

Census tracts for establishments were determined using the 2010 TIGER/Line shapefiles produced by the United States Census Bureau (United States Census Bureau, 2010).

Census tract population figures for 2000-2010 were taken from Neighborhood Socioeconomic and Demographic Characteristics of Census Tracts, United States, 2000-2010 (Clarke & Melendez, 2019). Population figures for 2011-2017 were obtained from Socioeconomic Status and Demographic Characteristics of Census Tracts, United States, 2008-2017 (Melendez et al., 2020).

Coverage

The dataset contains one observation per census tract per year in the United States, including Puerto Rico and the US Virgin Islands. Other US island territories are excluded.

Methodology

This dataset highlights the concentration of certain types of retail establishments per census tract in the United States. It focuses on shopping destinations such as stores selling furniture, clothing, books, music, building and garden supplies, and general merchandise. The number and density of these types of businesses has been in decline in recent years (Finlay et al., 2019,

Talen & Jeong, 2019), leading some to describe the sharp change in the retail environment as a “retail apocalypse” (Tokosh, 2018). Prior research has demonstrated that this change has implications not only for economic development and community planning, but also for public health. Retail businesses, along with more common gathering places (e.g. coffee shops, churches, gyms, senior centers), function as common “third places” in the United States. Third places are locations other than home (first place) and work/school (second place) where people gather, spend time, and casually interact with one another in ways that foster social interaction and community building. They are social infrastructure vital to civic life (Klinenberg, 2018), and their decline affects individual and community well-being (Finlay et al., 2018).

Our research team identified the following NAICS codes of interest:

- 442: Furniture and home furnishing stores.
- 443: Electronics and appliance stores.
- 444: Building material and garden supply stores, such as hardware and paint stores.
- 448: Clothing and accessory stores (excluding used clothing stores).
- 451: Sporting goods, hobby, musical instrument, and book stores, including yarn shops, toy stores, and newsstands.
- 452210: Department stores, which are defined as stores selling “general lines of new merchandise, such as apparel, jewelry, home furnishings, and toys, with no one merchandise line predominating” (United States Census Bureau, 2017). Note that this does not include warehouse clubs, which are not in this dataset but can be found in the NaNDA dataset [Grocery stores by census tract, United States, 2003-2017](#) (Finlay et al., 2020b).
- 453910: Pet and pet food stores.
- 453310: Used merchandise stores, such as those selling secondhand clothing, furniture, and antiques.
- 446120: Cosmetics, beauty supply, and perfume stores.
- 446191: Food (health) supplement stores, i.e. stores selling vitamins and nutritional supplements.

(This list is not inclusive of all retail businesses. See the usage note below for related datasets which, in combination with this one, provide a broader view of a neighborhood’s retail environment.)

For each NAICS code we identified, we queried the NETS database for all establishments that were open at some time between 2003 and 2015 and in 2017 (NETS does not contain data for 2016), and their location during each year. Note that NETS data were not cleaned or verified during this process, and that NETS data has limitations and inaccuracies as described in Gomez-Lopez et al. (2017) and Finlay et al. (2019).

We determined each establishment’s census tract for each year by mapping its latitude and longitude provided in the NETS database to the 2010 version of the TIGER/Line shapefiles from the US Census Bureau.

Using these locations, we created three variables to represent establishment counts per tract. The first count variable is the total of all establishments with that NAICS code in the tract. The second is the count of just establishments with non-zero sales figures in the current year, which can be used where desired to filter out establishments with very low business volume. The third is the count of establishments employing more than one person, which can be used to filter out very small owner-operated businesses if desired. The most useful variable for a given research question may vary by establishment type, so we have created three count types for all NAICS codes so that users may make their own decision about the version that best meets their needs.

We then calculated population densities (per 1000 people) for each of the three count variables for each NAICS code. Tract populations for 2003-2009 come from the NaNDA dataset [Neighborhood Socioeconomic and Demographic Characteristics of Census Tracts, United States, 2000-2010](#). Values for each year are interpolated. For 2010-2012 and 2013-2017, we used the tract population from the NaNDA dataset [Socioeconomic Status and Demographic Characteristics of Census Tracts, United States, 2008-2017](#). Population figures in this dataset were taken from the 2012 and 2017 ACS five-year estimates, respectively, in order to avoid overlapping ACS estimates as advised by the Census Bureau, and for compatibility with other NaNDA datasets. Population figures for 2010-2017 are not interpolated. Consult the codebooks for the NaNDA datasets Neighborhood Socioeconomic and Demographic Characteristics of Census Tracts, United States, 2000-2010 (Clarke & Melendez, 2019) and Socioeconomic Status and Demographic Characteristics of Census Tracts, United States, 2008-2017 (Melendez et al., 2020) for details on why these population figures were selected and how 2003-2009 population was interpolated.

We also created three variables representing the density of establishments per square mile. The area is the land area of the tract from the 2010 version of the TIGER/Line shapefiles. As a result, all area densities are standardized to 2010 census geography and are comparable across years across the entire dataset.

Population density was set to -1 for tracts with zero population in a given year. Population and area densities are missing for tracts in Puerto Rico and the US Virgin Islands, which are not included in the TIGER/Line shapefiles. Population and area densities are also missing in certain years for a few census tracts whose FIPS codes changed at some point between 2003 and 2017.

Finally, we merged counts and densities of all selected NAICS codes to create an aggregate dataset containing tract population, area, and count and density variables for all retail establishments per tract in each year.

Usage Note

Data users interested in the decline or expansion of segments of the retail environment might find further useful data in these additional NaNDA datasets:

- [Dollar stores by census tract, United States, 2003-2017](#) (Gomez-Lopez et al., 2020).

- [Eating and drinking places by census tract, United States, 2003-2017](#) (Esposito et al., 2020a), which includes coffee shops, fast food restaurants, and bars.
- [Grocery stores by census tract, United States, 2003-2017](#) (Finlay et al., 2020b), which includes warehouse clubs.
- [Health care services by census tract, United States, 2003-2017](#) (Khan et al., 2020), which includes pharmacies and stores selling optical goods and health supplies.
- [Liquor, tobacco, and convenience stores by census tract, United States, 2003-2017](#) (Finlay et al., 2020c).
- [Personal care services and laundromats by census tract, United States, 2003-2017](#) (Esposito et al., 2020b).
- [Post offices and banks by census tract, United States, 2003-2017](#) (Finlay et al., 2020d).

Those interested in third places and community resources might find further useful data in these additional NaNDA datasets which are not focused on the retail environment:

- [Arts, entertainment, and recreation organizations by census tract, United States, 2003-2017](#) (Finlay et al., 2020a), which includes gyms, libraries, and arcades, among others.
- [Religious, civic, and social organizations by census tract, United States, 2003-2017](#) (Finlay et al., 2020e).
- [Social services by census tract, United States, 2003-2017](#) (Finlay et al., 2020f), which includes senior centers.

Variables

For additional information on the definition of each NAICS code, see the Methodology section of this document and the 2017 NAICS website: <https://www.census.gov/cgi-bin/sssd/naics/naicsrch?chart=2017>.

Variable	Type	Obs	Unique	Mean	Min	Max	Label
tract_fips10	string	1036462	74033	.	.	.	Census tract FIPS code (2010 TIGER/Line shapefiles)
year	int	1036462	14	2009.571	2003	2017	Year
population	float	1020881	250271	4208.894	0	65528	Tract population
aland10	float	1022434	72534	48.07265	0	85425.73	Tract land area, square miles
count_442	int	1036462	121	2.316789	0	156	# furniture stores
count_sales_442	int	1036462	112	2.126896	0	148	# furniture stores w/ sales>0
count_emp_442	int	1036462	105	1.772615	0	128	# furniture stores w/ 2+ employees
popden_442	float	1015575	278208	2.471561	-1	289507.3	# furniture stores per 1000 people
popden_sales_442	float	1015554	268728	2.262326	-1	289507.3	# furniture stores per 1000 people w/ sales>0
popden_emp_442	float	1015541	247564	2.269165	-1	289507.3	# furniture stores per 1000 people w/ 2+ employees
aden_442	float	1017996	192089	2.654746	0	1443.139	# furniture stores per sq mile
aden_sales_442	float	1017996	186264	2.428714	0	1398.506	# furniture stores per sq mile w/ sales>0
aden_emp_442	float	1017996	171228	2.061152	0	1145.585	# furniture stores per sq mile w/ 2+ employees
count_443	int	1036462	122	2.273187	0	202	# electronics & appliance stores
count_sales_443	int	1036462	115	2.09335	0	181	# electronics & appliance stores w/ sales>0
count_emp_443	int	1036462	119	1.910329	0	182	# electronics & appliance stores w/ 2+ employees
popden_443	float	1015641	282654	44.64526	-1	1.17E+07	# electronics & appliance stores per 1000 people
popden_sales_443	float	1015605	273215	42.06461	-1	1.17E+07	# electronics & appliance stores per 1000 people w/ sales>0
popden_emp_443	float	1015622	260304	41.93649	-1	1.17E+07	# electronics & appliance stores per 1000 people w/ 2+ employees
aden_443	float	1017996	197955	3.061896	0	880.0734	# electronics & appliance stores per sq mile
aden_sales_443	float	1017996	191924	2.806688	0	868.7903	# electronics & appliance stores per sq mile w/ sales>0
aden_emp_443	float	1017996	183149	2.603726	0	868.7903	# electronics & appliance stores per sq mile w/ 2+ employees
count_444	int	1036462	76	2.24617	0	85	# building & garden supply stores

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Variable	Type	Obs	Unique	Mean	Min	Max	Label
count_sales_444	int	1036462	69	2.08469	0	75	# building & garden supply stores w/ sales>0
count_emp_444	int	1036462	70	1.890555	0	79	# building & garden supply stores w/ 2+ employees
popden_444	float	1015567	280424	3.989635	-1	579014.6	# building & garden supply stores per 1000 people
popden_sales_444	float	1015541	271725	3.743876	-1	579014.6	# building & garden supply stores per 1000 people w/ sales>0
popden_emp_444	float	1015542	261022	3.762698	-1	579014.6	# building & garden supply stores per 1000 people w/ 2+ employees
aden_444	float	1017996	185929	1.764495	0	265.8204	# building & garden supply stores per sq mile
aden_sales_444	float	1017996	180215	1.615567	0	236.9428	# building & garden supply stores per sq mile w/ sales>0
aden_emp_444	float	1017996	173210	1.490022	0	265.8204	# building & garden supply stores per sq mile w/ 2+ employees
count_448	int	1036462	354	4.784363	0	1296	# clothing stores
count_sales_448	int	1036462	340	4.371346	0	1158	# clothing stores w/ sales>0
count_emp_448	int	1036462	321	3.977495	0	1164	# clothing stores w/ 2+ employees
popden_448	float	1015782	340491	5.242418	-1	1158029	# clothing stores per 1000 people
popden_sales_448	float	1015750	329208	4.703522	-1	1158029	# clothing stores per 1000 people w/ sales>0
popden_emp_448	float	1015699	310732	4.568573	-1	868522.1	# clothing stores per 1000 people w/ 2+ employees
aden_448	float	1017996	270762	8.782971	0	14550.58	# clothing stores per sq mile
aden_sales_448	float	1017996	261738	8.027018	0	13841.51	# clothing stores per sq mile w/ sales>0
aden_emp_448	float	1017996	249556	7.234064	0	9838.259	# clothing stores per sq mile w/ 2+ employees
count_451	int	1036462	101	2.687595	0	152	# sporting goods, hobby, book, music stores
count_sales_451	int	1036462	90	2.496032	0	138	# sporting goods, hobby, book, music stores w/ sales>0
count_emp_451	int	1036462	67	1.618895	0	94	# sporting goods, hobby, book, music stores w/ 2+ employees
popden_451	float	1015679	303492	4.164237	-1	579014.6	# sporting goods, hobby, book, music stores per 1000 people
popden_sales_451	float	1015653	295529	3.797803	-1	579014.6	# sporting goods, hobby, book, music stores per 1000 people w/ sales>0
popden_emp_451	float	1015563	247679	2.695224	-1	289507.3	# sporting goods, hobby, book, music stores per 1000 people w/ 2+ employees
aden_451	float	1017996	215261	2.779009	0	1296.903	# sporting goods, hobby, book, music stores per sq mile
aden_sales_451	float	1017996	210486	2.579816	0	1194.338	# sporting goods, hobby, book, music stores per sq mile w/ sales>0
aden_emp_451	float	1017996	172951	1.738461	0	796.2252	# sporting goods, hobby, book, music stores per sq mile w/ 2+ employees

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Variable	Type	Obs	Unique	Mean	Min	Max	Label
count_452210	int	1036462	39	0.4002974	0	55	# department stores
count_sales_452210	int	1036462	39	0.372003	0	48	# department stores w/ sales>0
count_emp_452210	int	1036462	39	0.3861676	0	55	# department stores w/ 2+ employees
popden_452210	float	1015249	97874	0.9391934	-1	289507.3	# department stores per 1000 people
popden_sales_452210	float	1015237	92502	0.7491754	-1	289507.3	# department stores per 1000 people w/ sales>0
popden_emp_452210	float	1015249	94739	0.9347812	-1	289507.3	# department stores per 1000 people w/ 2+ employees
aden_452210	float	1017996	41352	0.4489252	0	230.8943	# department stores per sq mile
aden_sales_452210	float	1017996	40668	0.4104578	0	184.7155	# department stores per sq mile w/ sales>0
aden_emp_452210	float	1017996	40404	0.4255772	0	230.8943	# department stores per sq mile w/ 2+ employees
count_453910	int	1036462	12	0.3877856	0	11	# pet/pet food stores
count_sales_453910	int	1036462	11	0.3515672	0	10	# pet/pet food stores w/ sales>0
count_emp_453910	int	1036462	10	0.2639026	0	9	# pet/pet food stores w/ 2+ employees
popden_453910	float	1015209	107649	0.1238836	-1	3053.714	# pet/pet food stores per 1000 people
popden_sales_453910	float	1015203	100440	0.1119905	-1	3053.714	# pet/pet food stores per 1000 people w/ sales>0
popden_emp_453910	float	1015176	85284	0.0916087	-1	3053.714	# pet/pet food stores per 1000 people w/ 2+ employees
aden_453910	float	1017996	51094	0.3980145	0	131.9319	# pet/pet food stores per sq mile
aden_sales_453910	float	1017996	49961	0.3624684	0	98.94893	# pet/pet food stores per sq mile w/ sales>0
aden_emp_453910	float	1017996	44419	0.2759385	0	78.98094	# pet/pet food stores per sq mile w/ 2+ employees
count_453310	int	1036462	63	1.104914	0	71	# used merchandise stores
count_sales_453310	int	1036462	60	1.035775	0	70	# used merchandise stores w/ sales>0
count_emp_453310	int	1036462	43	0.5606081	0	45	# used merchandise stores w/ 2+ employees
popden_453310	float	1015236	206585	0.3926752	-1	10064.75	# used merchandise stores per 1000 people
popden_sales_453310	float	1015224	199914	0.3678691	-1	10064.75	# used merchandise stores per 1000 people w/ sales>0
popden_emp_453310	float	1015176	139548	0.2251542	-1	10064.75	# used merchandise stores per 1000 people w/ 2+ employees
aden_453310	float	1017996	113364	1.159652	0	1007.397	# used merchandise stores per sq mile
aden_sales_453310	float	1017996	110829	1.093469	0	993.2079	# used merchandise stores per sq mile w/ sales>0
aden_emp_453310	float	1017996	80932	0.6074073	0	638.4908	# used merchandise stores per sq mile w/ 2+ employees
count_446120	int	1036462	40	0.3854111	0	43	# cosmetics stores
count_sales_446120	int	1036462	37	0.3517621	0	40	# cosmetics stores w/ sales>0
count_emp_446120	int	1036462	28	0.239948	0	31	# cosmetics stores w/ 2+ employees

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Variable	Type	Obs	Unique	Mean	Min	Max	Label
popden_446120	float	1015251	101361	0.3532831	-1	119069	# cosmetics stores per 1000 people
popden_sales_446120	float	1015238	96675	0.2834105	-1	119069	# cosmetics stores per 1000 people w/ sales>0
popden_emp_446120	float	1015203	73278	0.1929048	-1	59534.49	# cosmetics stores per 1000 people w/ 2+ employees
aden_446120	float	1017996	48913	0.5652423	0	639.0482	# cosmetics stores per sq mile
aden_sales_446120	float	1017996	47968	0.5159679	0	594.4635	# cosmetics stores per sq mile w/ sales>0
aden_emp_446120	float	1017996	39709	0.3792096	0	416.1245	# cosmetics stores per sq mile w/ 2+ employees
count_446191	int	1036462	26	0.369728	0	25	# food/health supplement stores
count_sales_446191	int	1036462	24	0.3439094	0	23	# food/health supplement stores w/ sales>0
count_emp_446191	int	1036462	21	0.2597693	0	21	# food/health supplement stores w/ 2+ employees
popden_446191	float	1015185	108546	0.169054	-1	10064.75	# food/health supplement stores per 1000 people
popden_sales_446191	float	1015175	103815	0.1565629	-1	10064.75	# food/health supplement stores per 1000 people w/ sales>0
popden_emp_446191	float	1015158	86364	0.1283522	-1	10064.75	# food/health supplement stores per 1000 people w/ 2+ employees
aden_446191	float	1017996	45817	0.4767018	0	312.0844	# food/health supplement stores per sq mile
aden_sales_446191	float	1017996	45062	0.4420993	0	312.0844	# food/health supplement stores per sq mile w/ sales>0
aden_emp_446191	float	1017996	39195	0.343327	0	312.0844	# food/health supplement stores per sq mile w/ 2+ employees

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